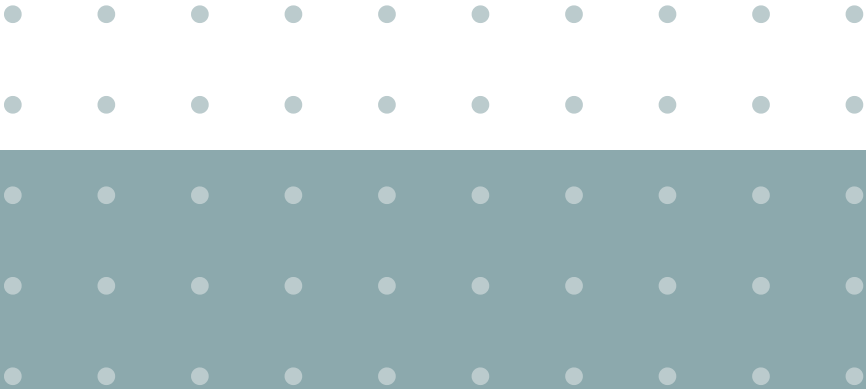




COURSE INTRODUCTION

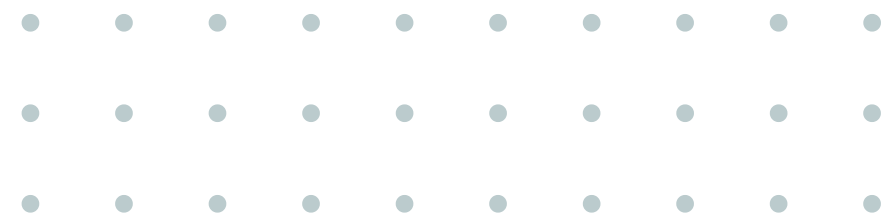
Dr. Sirasit Lochanachit

06026213 BIG DATA SYSTEMS



01. COURSE OVERVIEW

02. INTRODUCTION TO BIG DATA



**TABLE OF
CONTENT**



01.

COURSE OVERVIEW

06026213 BIG DATA SYSTEMS

COURSE REQUIREMENTS

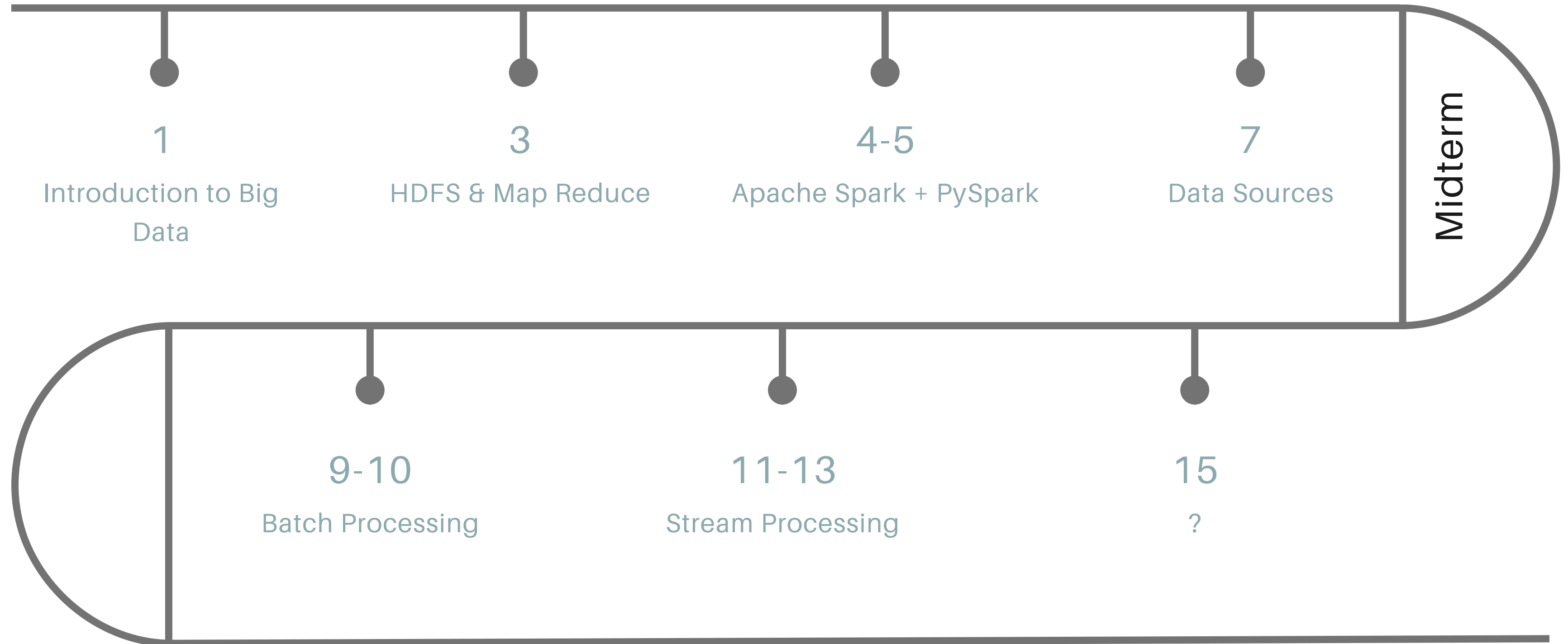
Able to do all of the followings:

- *Python Programming*
- *SQL Query*

COURSE DESCRIPTION

- *Cloud Installation? development?
maintenance? and management of Big Data.*
- *Basic Data Analysis on cluster using*
 - *SQL*
 - *MapReduce?*
 - *in-memory processing*
- *Data Analysis on*
 - *Structured data*
 - *Unstructured data?*
- *Big Data on cloud computing.*

Course Plan



GRADING

5% Attendance

45% Midterm Exam

50% Final Exam

ATTENDANCE

Out of 12 times, if you attend onsite

- *11+ times = 5%*
- *10 times = 4%*
- *9 times = 2%*
- *8 times = Unable to take a final exam.*

Note: Leave notice must be submitted at least 48 hrs before the lecture.

02.

INTRODUCTION TO BIG DATA

06026213 BIG DATA SYSTEMS



**“LIFE HAS TURNED INTO ONE
LONG STREAM OF NUMBERS
— EVEN WHEN WE’RE NOT
LOOKING”**

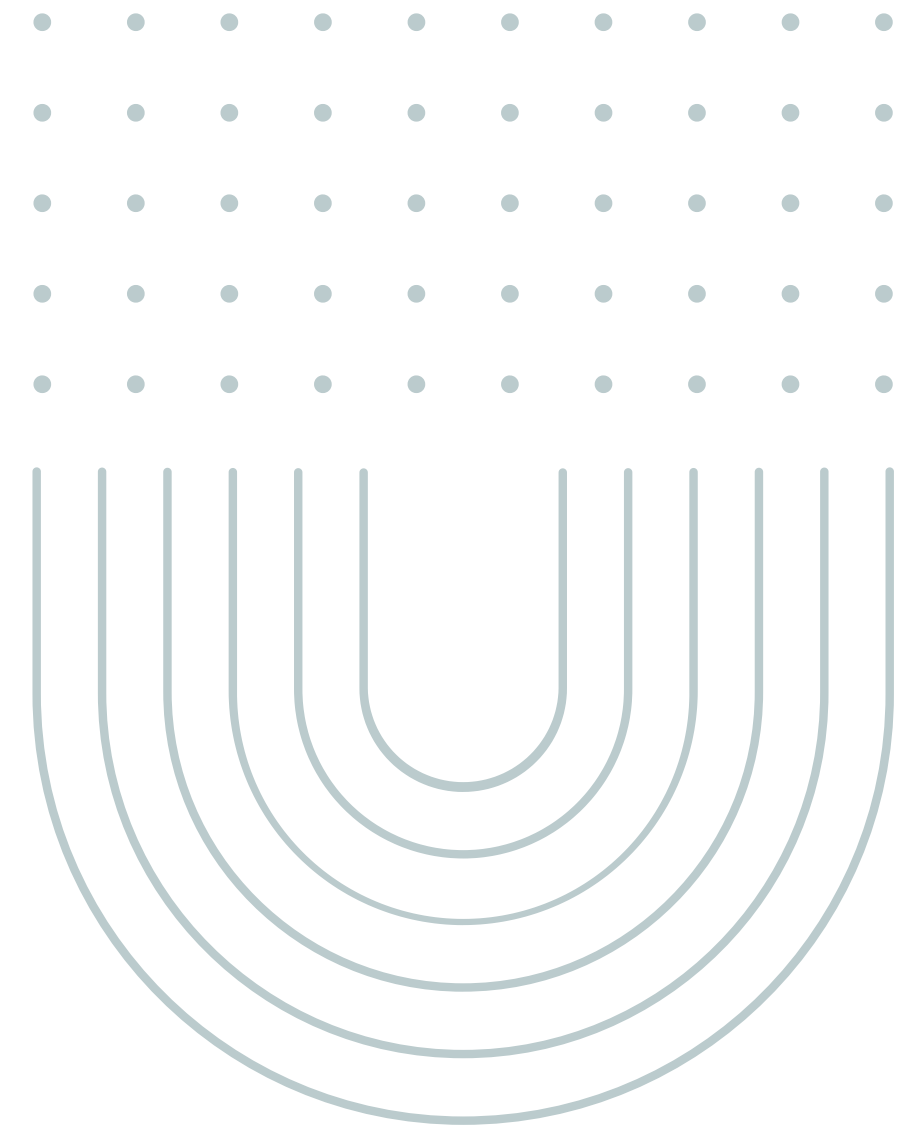




Browse the internet



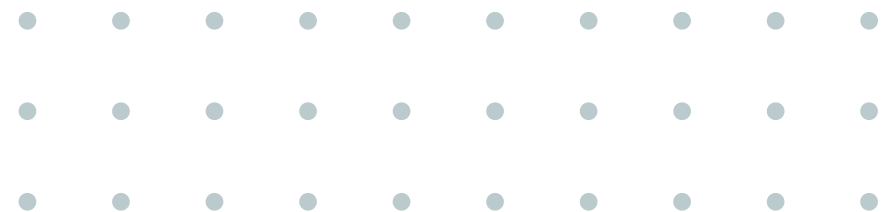
Text someone or post
something on social media



WHAT HAVE
YOU DONE
TODAY?

WHAT IS BIG DATA?

The analysis, processing, and storage of **large collections of data** that frequently originate from **disparate sources**

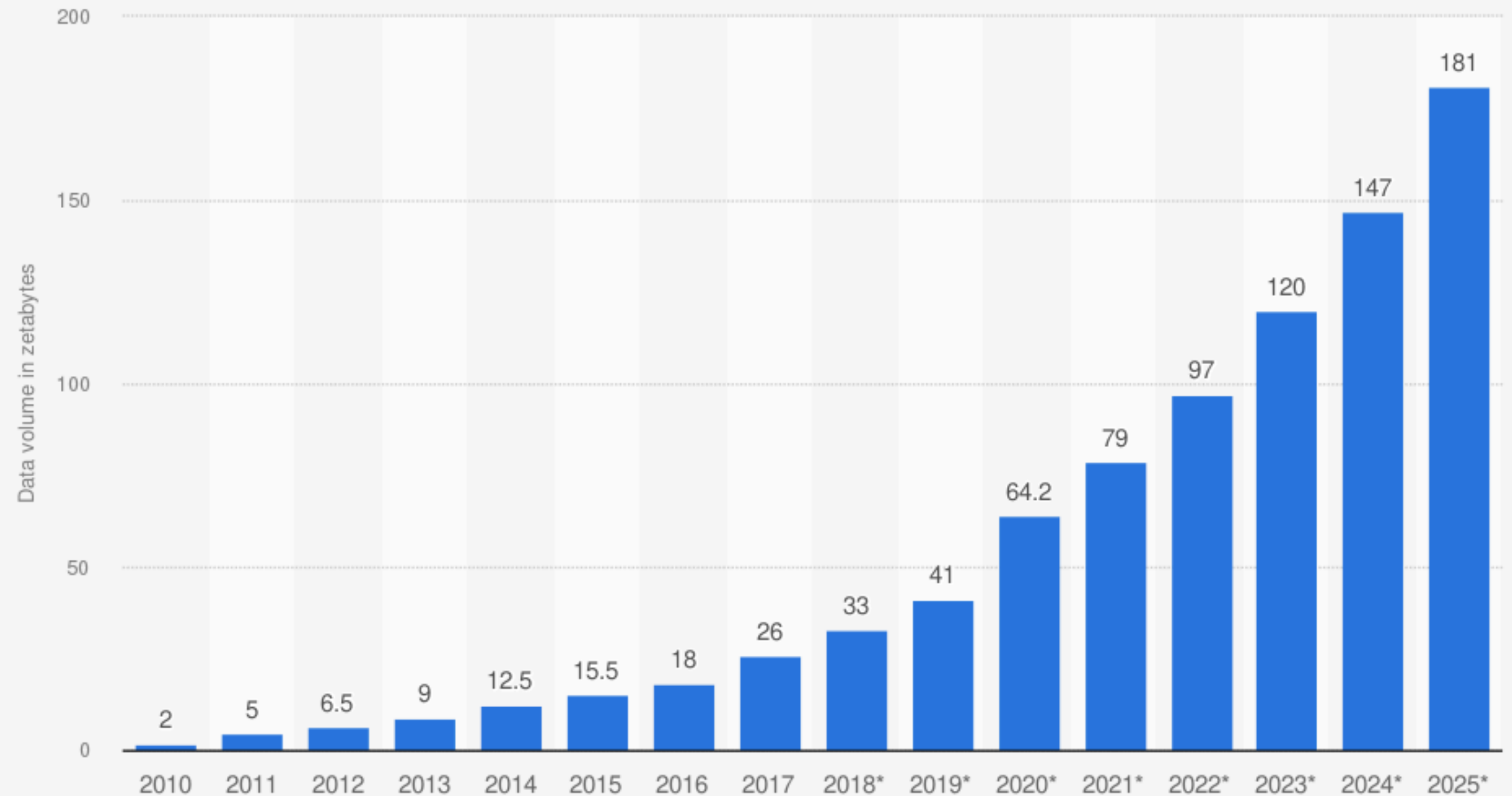


ORIGINAL CONCEPT: 3V

- Volume: The amount of data being generated
 - Online transaction
 - Health records
 - Sensors



Volume of data/information created, captured, copied, and consumed worldwide from 2010 to 2020, with forecasts from 2021 to 2025 (in zettabytes)



Sources
IDC; Seagate; Statista estimates
© Statista 2024

Additional Information:
Worldwide; 2010 to 2020

ORIGINAL CONCEPT: 3V

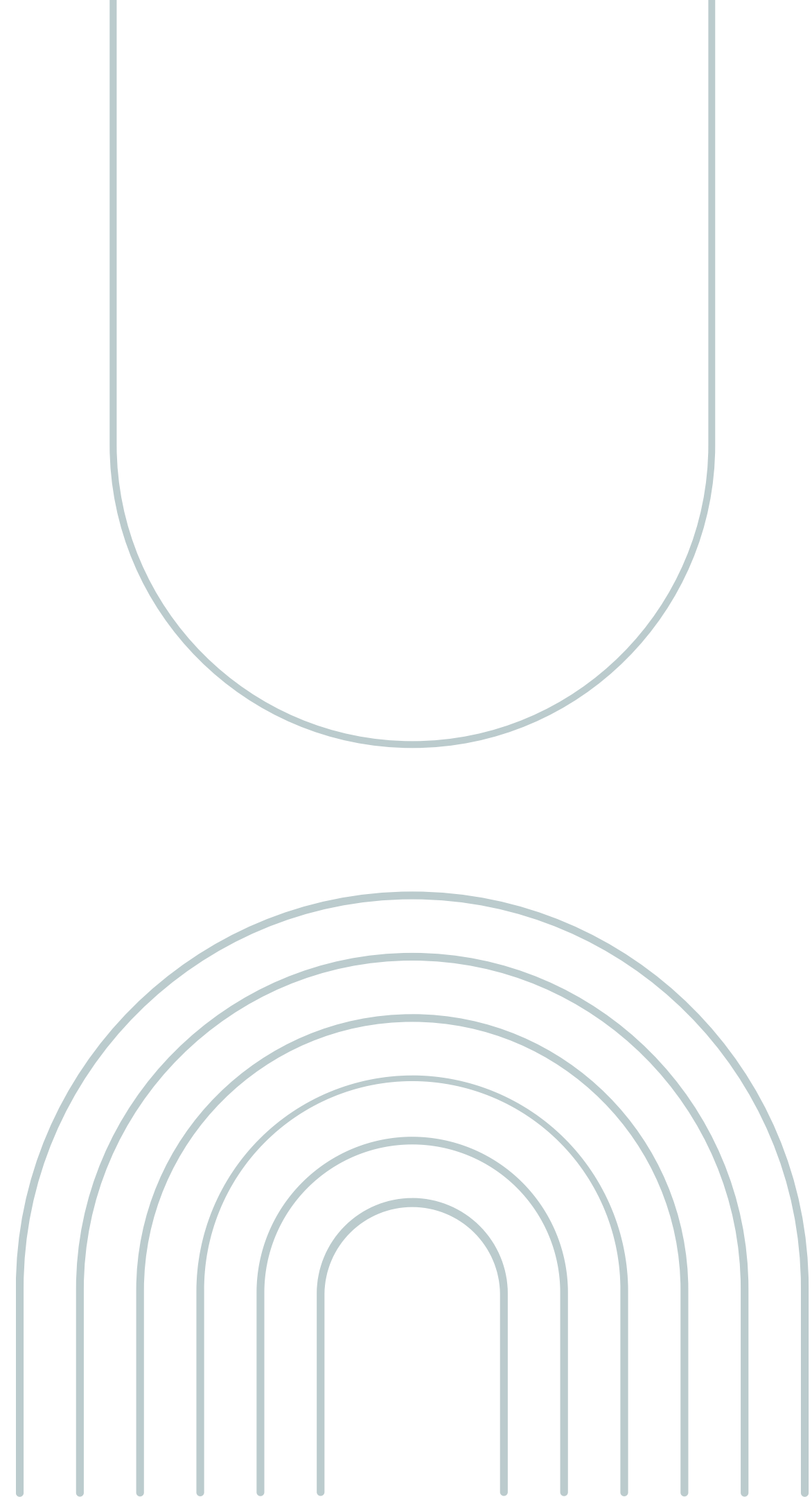
- Variety: The different types of data generated
 - Structured
 - Unstructured
 - Semi-structured

A collection of data formats and types arranged in a grid-like pattern. The items are as follows:

Row	Column 1	Column 2	Column 3	Column 4
1	Excel	CSV	JSON	DataFrame
2	Word (file)	Image	HTML	MySQL
3	Video	Audio	Sensor data	
4	PDF		Social media post	
5	XML	Text data		
6	E-mail		Server logs	Graph data

ORIGINAL CONCEPT: 3V

- Velocity: The speed at which data is generated
 - Batch
 - Data Stream
 - Near real-time
 - Real-time

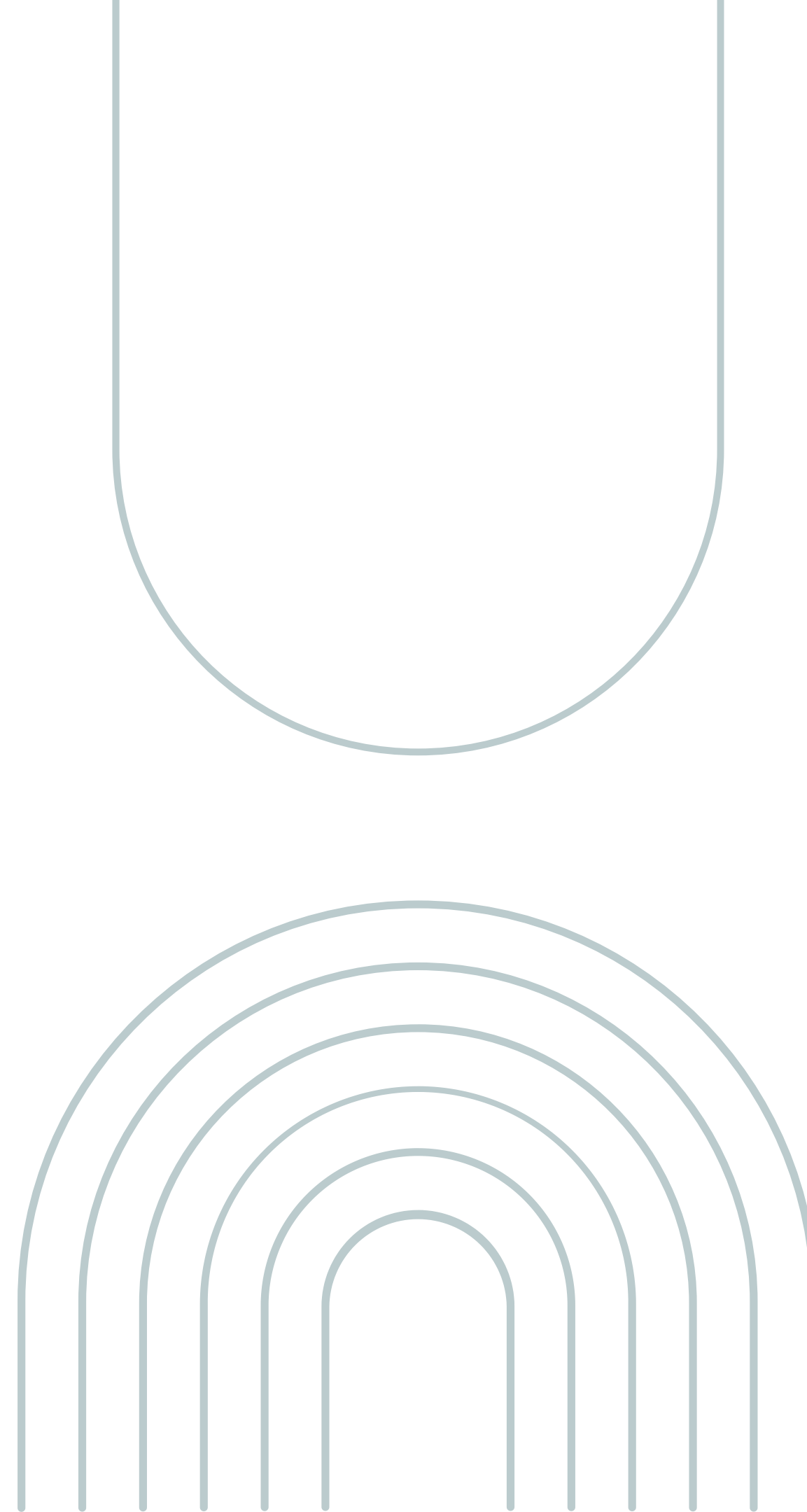


The Internet in Real-Time

How Quickly Data is Generated



By the way, in the 9 seconds you've been on this page, approximately 203166 GB of data was transferred over the internet.



ADDITIONAL CONCEPTS

- Veracity: The truthfulness or reliability of the data
 - High: Trustworthy and Accurate
 - Low:
 - Noise
 - Error
 - Ambiguity



ADDITIONAL CONCEPTS

- Value
 - Usefulness
 - Insights
 - Drives decisions and create benefits

ADDITIONAL CONCEPTS

- Variability
- Visualization
- Viscosity
- Virality
- and more



“What are the differences between data warehouse and big data system?”

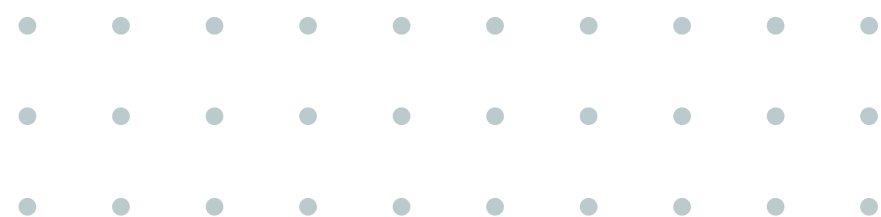
DATA WAREHOUSE (DW)

- Centralized repository for **structured data** collected from multiple sources (homogeneous)
- ETL
- Relies on relation databases with schemas (SQL-based)



LIMITATIONS OF DW

- Scalability Challenges
- Lack of Support for Unstructured and Semi-structured data
- Inflexible Schema Design
- Limited Real-time Processing



KEY DIFFERENCES

DATA WAREHOUSE

BIG DATA SYSTEM

Purpose

Data Velocity

Data Variety

Scalability